PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Indicators of Response to Exercise Training: A Systematic Review & Meta-Analysis
AUTHORS	Ardavani, Arash; Aziz, Hariz; Phillips, Bethan; Doleman, Brett; ramzan, imran; mozaffar, boshra; Atherton, Philip; Idris, Iskandar

VERSION 1 – REVIEW

REVIEWER	Dankel, Scott Rowan University
REVIEW RETURNED	03-Dec-2020

GENERAL COMMENTS	The authors look at the efficacy of endurance exercise, HIIT, and resistance exercise and compare whether VO2 max is increased with each mode of exercise above that of a control. They also compare the efficacy of endurance exercise compared to that of HIIT and resistance exercise. Finally, they include a comparison of different exercise responses in relation to that of VO2 max. The manuscript is well-written but I have several major concerns with the manuscript as follows: I really don't see the benefit of the dichotomous data approach for assessing differeces in responders. If there are significant differences between the exercise and control groups, don't we already know that the exercise group will have more responders than the control group (assuming the analyses are appropriately computed on somewhat normally distributed data)? Nonetheless, the idea that using each individual studies definition of individual responders seems very problematic as the results of the meta-analysis will be driven by which approach is taken. For example, if anything above 0 is considered a favorably response, even 50% of the control group will demonstrate a favorable response by definition (as the likelihood of obtaining the exact pre and post test is essentially 0). Whereas individual considering anything above the minimal difference will see much fewer responders. As such, in Figure 2A, the results of the analysis do not make sense. The OR are 0 and there is no variability. This is a product of how the individual responders were classified. Why not use the SD of the control group to determine the proportion of individual responders as depicted elsewhere (https://pubmed.ncbi.nlm.nih.gov/25823596/). As the authors describe in this paper each of these studies are not making appropriate analyses if the comparisons and proportion of differential responders is not made relative to a control group. Thus, the present study examines differential responders using methods not appropriate to classify differential responders. More information needs to be included in the analysis. For example,

	are the effect sizes in absolute units, standardized units, etc. And what is the variability used, it is the pre-test variability, change score variability, etc. I don't have a good idea of what I am interpreting with the analysis. The forest plots should be adjusted. Some of the forest plots have too small of an x axis range and the data cannot be seen (Figure 2A). Other times the x axis range is too large and the effect cannot be seen and interpreted (Figure 2B). Many of the mean weighted effect sizes are computed off 2 effect sizes. It makes more sense just to interpret these two studies than to include them in part of a meta-analysis as the point estimate will be very unstable (as shown in the figures). What is the significance of the colors used in the figures?

REVIEWER	Gavin, James University of Southampton, School of Health Sciences
REVIEW RETURNED	07-Dec-2020

GENERAL COMMENTS	**General comments** The submitted manuscript reports a systematic review and meta-analysis to investigate VO2max, as a means of measuring exercise training response. The authors provide three clear objectives (training modality; END vs HIIT/RET; VO2max vs other response measures), which are consistent throughout. The methods are sound in repeatability and results detailed in statistics, yet the Discussion could further expand upon the findings relating to Power Output as a viable measure to assess exercise training response. The paper is suitable to the journal's readership and high-quality in content and supplementary evidence. My specific comment can be found below. **Specific comments** **Abstract** • Insightful and appropriate. • Consider stating how many papers were initially identified, alongside the 29 selected. **Methods** The Method is logical in structure and content, particularly for the systematic review analyses. However: • Consider providing brief rationale for selecting only RCTs, case-control and cohort studies. • Consider providing an example of dichotomous and continuous data to illustrate your point. • Although you state 'grey area sources', it is not clear what grey literature you actually used. Presumably, and from reading the PRISMA diagram, it is PubMed and google, but be explicit. • With regards to 'Patient and public involvement' were any groups or stakeholders involved in the inception of the idea, planning and/or interpretation of the findings? Specifically, those

	outside the research team/authors. If so, I would briefly add. **Results** • Little detail for reasons for exclusions on your PRISMA diagram. **Discussion** • The discussion provides insightful review relating to the three objectives. However, I feel there needs to be further discussion on 'the apparent role for PO as a potential alternative' and greater review of the study's potential implications and/or recommendations arising. **Tables and figures** • Table 2 – weak title, please revise to add greater meaning (think PICO).

REVIEWER	Ikeda, Erika University of Cambridge, School of Sport and Recreation
REVIEW RETURNED	20-Jan-2021

GENERAL COMMENTS	General comments: Thank you for the opportunity for reviewing this paper. This systematic review aimed to investigate exercise response rates in the context of VO2 max. There is a concern about statistical methods used in this review. The authors need to clearly explain why dichotomous and continuous data were separately analysed using different methods (i.e., Mantel-Haenszel and inverse variance methods). There is a way to combine dichotomous and continuous outcomes to undertake meta-analyses (please see details here: https://handbook-5-1.cochrane.org/chapter_9/9_4_6_combining_dichotomous_and_continuous_outcomes.htm). In addition, it is highly recommend that standardised mean difference rather than mean difference is used for the outcome measures as it is likely that the outcomes were measured in various ways and scales can be different across studies. In this way, the authors will use only one statistical method – the generic inverse-variance method. Figures 2-5 need improvement – for example, all abbreviations should be included in footnote. It is assumed that these figures were copied from the RevMan, and there is a lack of explanation for readers to interpret the data (e.g., M.H. = Mantel-Haenszel method; IV = inverse-variance method). All figures (and tables) should stand-alone.

REVIEWER	Phan, Hoang University of Tasmania, Menzies Institute for Medical Research
REVIEW RETURNED	11-Feb-2021

GENERAL COMMENTS	This is an interesting SR/MA to investigate indicators of response to exercise Training. I have a few comments below: 1) Why the systematic review/MA by Gurd et al (2016) was included in the meta-analyses. It is more common to

	include data from original research in the MA. 2) I acknowledge that data-points were used in the meta-analyses, meaning different estimates from the same study are combined. In the method, you mentioned that random effects modes were performed. It is reasonable if there are differences between study design, population, etc. I therefore would suggest combining estimates of the same outcome using fixed effects approach first, then polling this with those from other studies. 3) There are too many figures (and subfigures) in the paper, which limits the accessibility for readers. I would suggest increase the font size of the figures. It would be better to include outcomes with a good number of studies (rather than only two studies). An alternative option is to include the estimates from only a few studies in the main text or supplemental materials. 4) In Figure 2B, 2C and 3B, data from 8 studies were pooled. Although meta-regression and publication bias have been suggested to perform among at least 10 studies. There is considerable heterogeneity between the studies included in the meta-analysis. As an ultimate goal of meta-analysis is to explore sources of heterogeneity, I would suggest to perform meta-regression or/and stratified meta-analytic summarization. This will allow further understand about if there are potential factors that affect the association between exposure and outcomes differently, rather than just stick to the highest standards of guidelines.

VERSION 1 – AUTHOR RESPONSE

Reviewer	Reviewer's Comments	Response	
1	The authors look at the efficacy of endurance exercise, HIIT, and resistance exercise and compare whether VO2 max is increased with each mode of exercise above that of a control. They also compare the efficacy of endurance exercise compared to that of HIIT and resistance exercise. Finally, they include a comparison of different exercise responses in relation to that of VO2 max. The manuscript is well-written but I have several major concerns with the manuscript as follows:		
	I really don't see the benefit of the dichotomous data approach for assessing differences in responders. If there are significant differences between the exercise and control groups, don't we already know that the exercise group will have more responders than the control group (assuming the analyses are appropriately computed on somewhat normally distributed data)? Nonetheless, the idea that using each individual studies definition of individual	Thank you for your highly instructive feedback. We agree with this reviewers comments. We have therefore reconsidered our incorporation of dichotomous data in view of the points raised and have decided to remove the dichotomous data-based	Multiple deletions, majority of 'Results' modified, half of 'Discussion' modified.

	responders seems very problematic as the results of the meta-analysis will be driven by which approach is taken. For example, if anything above 0 is considered a favorably response, even 50% of the control group will demonstrate a favorable response by definition (as the likelihood of obtaining the exact pre and post test is essentially 0). Whereas individual considering anything above the minimal difference will see much fewer responders. As such, in Figure 2A, the results of the analysis do not make sense. The OR are 0 and there is no variability. This is a product of how the individual responders were classified. Why not use the SD of the control group to determine the proportion of individual responders as depicted elsewhere (https://pubmed.ncbi.nlm.nih.gov/25823596/). As the authors describe in this paper each of these studies are not making appropriate analyses if the comparisons and proportion of differential responders is not made relative to a control group. Thus, the present study examines differential responders using methods not appropriate to classify differential responders.	analyses from the study. Further, we attempted the SD-based intervention effect estimation as advised. Unfortunately, several studies either did not contain a control group to permit the undertaking of an SD-based estimation (e.g. Bonafiglia et al. 2016, Ross et al. 2015) or did not contain pre- and post-intervention SD data for both controls and intervention groups (e.g. Earnest et al. 2011, Pandey et al. 2015). Instead, where possible, those studies which were originally taken as sources for dichotomous data, but contained equivalent continuous data, were incorporated into the existing continuous data-based analysis where possible. Said analysis has been repeated and any changes to the results and analysis have been updated where appropriate. We have also removed any reference to dichotomous data within our manuscript and have removed all the forest plots and analyses based on this data. As such, the 'Domain A' and 'Domain B' designations are now redundant, and have been removed.	
	More information needs to be included in the analysis. For example, are the effect sizes in absolute units, standardized units, etc. And what is the variability used, it is the pre-test variability, change score variability, etc. I don't have a good idea of what I am interpreting with the analysis.	We have addressed these inadvertent omissions in our 'Methods' section. In addition, a residual from a prior draft version was corrected (inclusion criteria for the intervention was corrected from 'three' to 'two' weeks).	Page 10, line 291 Page 10, line 278
	The forest plots should be adjusted. Some of the forest plots have too small of an x axis range and the data cannot be seen (Figure 2A). Other times the x axis range is too large and the effect cannot be seen and interpreted (Figure 2B).	Thank you for highlighting this formatting issue – We have increased the size of the forest plots and have modified the x-axes where appropriate.	Figures 2-5
	Many of the mean weighted effect sizes are computed off 2 effect sizes. It makes more sense just to interpret these two studies than to include them in part of a meta-analysis as the point estimate will be very unstable (as shown in the figures).	Thank you for your suggestion, which we have applied through i) removing all forest plots consisting of only two studies and ii) stating the outcomes in the 'Results' and 'Discussion'	Page 14, line 400 to Page 15, line 457. Table 2 now removed.

		sections (where appropriate), and iii) removal of Table 2 (rank order comparison of forest plot-generated effect sizes) due to a paucity of data following our actions to address this comment.	
	What is the significance of the colors used in the figures?	The differing colours in the forest plots were a consequence of 'RevMan''s allocation secondary to the different data types used (dichotomous and continuous). As we have removed dichotomous data from our analysis per your first comment, the forest plots are now represented by a single colour.	N/A
2	Please see attached document: "Reviewer comments_bmjopen-2020-044676".		
	Consider stating how many papers were initially identified, alongside the 29 selected.	Thank you for highlighting this oversight – We have explicitly stated the number of papers that were initially identified in the Abstract.	Page 4, line 85
	Consider providing brief rationale for selecting only RCTs, case-control and cohort studies.	Thank you for your comment – We have provided rationale regarding the basis for our selection of the included study types in our manuscript.	Page 9, lines 244-246
	Consider providing an example of dichotomous and continuous data to illustrate your point.	We have provided an example as advised for continuous data. As a consequence of our change following the input of Reviewer #1, dichotomous data is no longer utilised in our investigation.	Page 9, line 240
	Although you state 'grey area sources', it is not clear what grey literature you actually used. Presumably, and from reading the PRISMA diagram, it is PubMed and google, but be explicit.	Thank you for highlighting this – We have explicitly stated which grey literature sources were cited within the manuscript.	Page 9, line 261
	With regards to 'Patient and public involvement' were any groups or stakeholders involved in the inception of the idea, planning and/or interpretation of the findings? Specifically, those outside the research team/authors. If so, I would briefly add.	Thank you for your comment – We have now specified that no external groups, stakeholders or members of the public were involved in any element of the study's inception, planning, implementation or analysis.	Page 12, line 336-340
	Little detail for reasons for exclusions on your PRISMA diagram	Thank you for indicating this oversight, which we have corrected, through the statement of the five most frequent reasons why other studies were excluded. Furthermore, we have replaced the existing Figure 1 with the PRISMA-approved equivalent to standardise the appearance.	Revised Figure 1

	The discussion provides insightful review relating to the three objectives. However, I feel there needs to be further discussion on 'the apparent role for PO as a potential alternative' and greater review of the study's potential implications and/or recommendations arising.	Thank you for this beneficial suggestion. Additional interpretations of this outcome's findings have been provided in the 'Discussion' section. Our previous findings concerning the potential role of PO as an alternative to VO2max have been modified accordingly, following the change in results downstream to the recommendations made by Reviewers 1 and 3. Further, we have updated our Strength and Limitations section in view of all Reviewer suggestions.	Page 16, line 466 – Page 19, line 569
	Table 2 – weak title, please revise to add greater meaning (think PICO).	Thank you – Following Reviewer #1's request, we have removed all forest plots calculated with only two datapoints. As such, the overall rationale of Table 2 was substantially reduced. We have therefore removed Table 2. We took this opportunity to review Table 1, and have made substantial changes in formatting to improve overall appearance and readability.	Revised Table 1
3	Thank you for the opportunity for reviewing this paper. This systematic review aimed to investigate exercise response rates in the context of VO2 max.		
	There is a concern about statistical methods used in this review. The authors need to clearly explain why dichotomous and continuous data were separately analysed using different methods (i.e., Mantel-Haenszel and inverse variance methods). There is a way to combine dichotomous and continuous outcomes to undertake meta-analyses (please see details here: https://handbook-5-1.cochrane.org/chapter_9/9_4_6_combining_dichotomous_and_continuous_outcomes.htm).	Thank you for your comment. This was highlighted in another reviewer's comments. Following Reviewer #1's request, we have indirectly resolved this concern, as we have removed the utilisation of dichotomous data, and have utilised SD-based data where available.	Multiple (per Reviewer #1 Comment #1)
	In addition, it is highly recommend that standardised mean difference rather than mean difference is used for the outcome measures as it is likely that the outcomes were measured in various ways and scales can be different across studies. In this way, the authors will use only one statistical method – the generic inverse-variance method.	Thank you for your helpful comment, which we have fully applied to our repeat attempt at analysing our existing data.	Page 11, line 311
	Figures 2-5 need improvement – for example, all abbreviations should be included in footnote. It is assumed that these figures were copied from the RevMan, and there is a lack	Thank you for highlighting this oversight – We have provided a full explanation of the abbreviations for each figure	Figures 2-5

	of explanation for readers to interpret the data (e.g., M.H. = Mantel-Haenszel method; IV = inverse-variance method). All figures (and tables) should stand-alone.	and table in a standalone manner as advised. An update of Abbreviations List (acronyms listed in alphabetical order; reformatted; additions made) was also undertaken to ease the interpretation and consistency of the manuscript.	
4	This is an interesting SR/MA to investigate indicators of response to exercise Training. I have a few comments below:		
	Why the systematic review/MA by Gurd et al (2016) was included in the meta-analyses. It is more common to include data from original research in the MA.	Thank you for your comment. Gurd et al. 2016 (DOI: 10.1139/apnm-2015-0449) is a pooled analysis of five other studies, where either the entire dataset or a subset of the cohort were re-assessed with the addition of a measurement error (TE) parameter for cardiopulmonary testing (VO2max). The authors did not undertake a systematic assessment or conduct any weighting between the five populations. Nonetheless, we recognise the inclusion of this study without an explicit statement as to the justification for the inclusion of Gurd et al. (and not the individual studies) as an oversight and have addressed this accordingly within the manuscript.	Page 11, line 298-300
	I acknowledge that data-points were used in the meta-analyses, meaning different estimates from the same study are combined. In the method, you mentioned that random effects modes were performed. It is reasonable if there are differences between study design, population, etc. I therefore would suggest combining estimates of the same outcome using fixed effects approach first, then polling this with those from other studies.	Thank you for this comment, which addresses the datapoints used from several dichotomous forest plots. As we have removed these per Reviewer #1's request, a random effects model for the remaining analyses, which consist of continuous data overwhelmingly derived from individual datapoints sourced from different studies, would remain the appropriate choice given our prior modifications.	N/A
	There are too many figures (and subfigures) in the paper, which limits the accessibility for readers. I would suggest increase the font size of the figures. It would be better to include outcomes with a good number of studies (rather than only two studies). An alternative option is to include the estimates from only a few studies in the main text or supplemental materials.	Thank you for highlighting this limitation which has also been raised by another reviewer. In accordance with this and another reviewer's comments, the number of figures and subfigures has been reduced following our removal of the forest plots based on dichotomous data. We	Figures 2-5

		have also increased the size of the forest plots accordingly.	
	In Figure 2B, 2C and 3B, data from 8 studies were pooled. Although meta-regression and publication bias have been suggested to perform among at least 10 studies. There is considerable heterogeneity between the studies included in the meta-analysis. As an ultimate goal of meta-analysis is to explore sources of heterogeneity, I would suggest to perform meta-regression or/and stratified meta-analytic summarization. This will allow further understand about if there are potential factors that affect the association between exposure and outcomes differently, rather than just stick to the highest standards of guidelines.	Thank you for this excellent insight and suggestion. We agree a meta-regression would have robustly quantified the relationships of the studied exercise indicators with the studied exercise interventions. Unfortunately, less than ten studies at the synthesis stage contain all of the indicators of interest and the exercise interventions. As such, the available data does not permit a meta-regression analysis. We have nonetheless taken this recommendation and have suggested it as an area for further research. With respect to stratification of the meta-analyses, following a second review of the available data, we were able to undertake a stratified assessment based on intervention duration (less than 12 weeks, more than or equal to 12 weeks). We have therefore incorporated this summarily in our repeat analyses.	Page 11, line 324; Page 19, like 560-563

VERSION 2 – REVIEW

REVIEWER	Ikeda, Erika University of Cambridge, School of Sport and Recreation
REVIEW RETURNED	22-Mar-2021

GENERAL COMMENTS	The authors responded to the reviewer's comment appropriately and made adequate changes. No further comments are given.

REVIEWER	Phan, Hoang University of Tasmania, Menzies Institute for Medical Research
REVIEW RETURNED	05-Apr-2021

GENERAL COMMENTS	Thank you for addressing my comments.